Harshit Malla

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Education

McMaster University — Hamilton, ON *Sep 2025 – Apr 2029 (Expected)*

B.Eng., Computer Engineering & Management (Second Year) **GPA: 3.86 / 4.0**

- Awards: McMaster Award of Excellence ($3,000) · Alexander Rutherford Scholarship ($2,500) · YMM Rising Stars Award · CEMC Certificate of Distinction (Univ. of Waterloo) · ASAA Badminton Provincials Gold

Skills

- **Hardware & RTL** Arduino, soldering, wiring, circuit prototyping
- **Software:** Python, C/C++ (PlatformIO), MATLAB, Git/GitHub, REST APIs
- **Verification & Tools:** Autodesk Inventor (CAD)

Work Experience

VEX Robotics — Team Lead, Mechatronic Design & Systems Integration *Sep 2019 – Jun 2025*

- Led design, CAD modeling, and fabrication of competition robots across 6 seasons, ranking **165th of 20,000+ teams worldwide** in the 2024–25 VEX World Championship season
- Won the **VEX Robotics Alberta Provincial Championship** and the **Excellence Award** (Mar 2025), the program's highest combined award for performance, design documentation, and interviews
- Ran full design cycles in Autodesk Inventor — simulating drivetrain and lift mechanisms before fabrication to cut rebuild iterations and keep robots competition-ready across **10+ tournaments per season**
- Resolved internal and cross-team conflicts during alliance play, coordinating strategy with partner teams under match-time pressure

Projects

PHANTOM-16 — 16-Bit Pipelined RISC CPU *(RTL Design & Verification — Verilog, Icarus, FPGA)* *May 2026 – Present*

- Architecting a **5-stage pipelined (IF/ID/EX/MEM/WB) 16-bit RISC processor** from scratch in synthesizable Verilog for a **Sipeed Tang Nano 9K FPGA** (GW1NR-9C); authored a full PRD covering ISA, microarchitecture, hazard/forwarding scheme, and a verification plan targeting 100% statement coverage
- Designed and froze a custom **23-instruction ISA** across 3 fixed-width encoding formats (R/I/J) with an 8-register ABI (R0 hardwired) and a memory-mapped I/O region for UART/GPIO/7-seg peripherals
- Implemented and verified the **ALU** (9 operations, latch-free combinational, alu_op encoded to mirror R-type funct for zero-cost decode) and **8×16 register file** (dual read ports, R0-hardwired, write-after-read semantics chosen to match the planned MEM-WB forwarding path)
- Built a self-checking **Icarus Verilog testbench suite** with **20/20 unit tests passing** on the ALU and register file, covering edge cases including 16-bit overflow wrap, arithmetic-vs-logical right-shift sign preservation, and same-cycle read-during-write on the register file

PantryPal — AI Recipe-to-Cart Pipeline *(H0 Hackathon — Python, Vision AI, REST APIs)* *Jun 2026*

- Built an app that converts any recipe — URL, TikTok screenshot, or cookbook photo — into a store-matched, priced shopping list in **under 10 seconds**, removing manual transcription from the grocery workflow
- Engineered a dual-input pipeline (**schema.org JSON-LD parsing + vision-AI screenshot extraction**) converging into one 8-stage normalize→match→price flow, hitting **90%+ ingredient extraction accuracy** on recipe URLs
- Solved unit-to-package reconciliation (e.g., 2 tbsp → one 1 lb block) with a canonical ingredient dictionary and conversion layer, keeping cost estimates within **±15% of real checkout totals**
- Cut API spend with a **24-hour TTL product-lookup cache** and query deduplication; designed a no-scraping architecture using only user-supplied screenshots and publisher-exposed metadata

Assistive Grabber Arm — Electromechanical Client Design *(McMaster ENG 1P13 — CAD, Rapid Prototyping)* *Jan – Apr 2026*

- Designed and delivered a 3D-printed reach-and-grab device for a real wheelchair-user client, taking 3 concept designs through a weighted decision matrix to a working physical prototype in **13 weeks**
- Validated structural performance under load — main rod sustained **48 N** with the end effector gripping at **9.8 N**; redesigned the claw geometry to a **45° contact angle** after initial grip failures in testing
- Modeled all components in Autodesk Inventor and iterated through multiple PLA print revisions, presenting the final design and test data to the client and faculty panel

MACRO-TICK — Real-Time Embedded Trading Terminal *(ESP32-S3, C++, I2C, UART, Python)* *Feb 2026 – April 2026*

- Developed embedded firmware in C++ (PlatformIO) on an **ESP32-S3** driving SSD1306 OLED displays over a **100 kHz I2C bus** with 4.7 kΩ pull-ups, rendering live equity prices streamed over a custom **115200-baud UART protocol** from a Python host
- Designed a safety-critical **4-state hold-to-arm execution FSM** (idle → pressed → armed → cooldown) with a **200 ms arm threshold**, 2 s auto-disarm timeout, and 2 s cooldown — making accidental trade execution physically impossible from the hardware side
- Brought up and debugged the full hardware stack on breadboard — I2C bus stability, pull-up sizing, supply decoupling (100 µF) — and validated firmware against a **Python market-data simulator** before live brokerage API integration